

Student Name:

Student ID:

Habib University
Dhanani School of Science and Engineering
Computer Architecture
EE 371 / CS 330 / CE 321 (Fall 2025)

Quiz 3

Date: 26th March, 2025

Total marks: 20

Obtained Marks:

Total time: 50 mins

Instructions:

1. All questions should be answered **with ink pen**.
2. Smart watches, laptops, and similar electronics are **strictly NOT allowed**.
3. Answer sheets should contain all steps, working, explanations, and assumptions.
4. Write your name and HU ID on all sheets.
5. Turn in your question paper along with your answer sheets.
6. You are not allowed to ask/share your method or answer with your peers. The work submitted by you is solely your own work. Any violation of this will be the violation of HU Honor code and proper action will be taken as per university policy if found to be involved in such an activity.

CLO Assessment:

This quiz assesses students for the following course learning outcomes.

Course Learning Outcomes		CLO Assessed
CLO 1	<i>Explain</i> the role of ISA in modern processors and instruction encodings and assembly language programming	
CLO 2	<i>Explain</i> the architecture and working of a single cycle processor	
CLO 3	<i>Design</i> the architecture to mitigate issues of a pipelined processor	✓
CLO 4	<i>Analyze the</i> performance of cache operations	

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Q1: Let's say we have a 5-stage RISC-V processor. Delay for each stage is given below in a tabular form.

Fetch	Decode	Execute	Memory	Write-back
250	250	300	300	300

a) What will be clock cycle time for a non-pipelined version of this processor? **(1-point)**

Sol: 1400

b) What will be the clock cycle time for a pipelined version of this processor? **(1-point)**

Sol: 300

c) What is the maximum speed-up we can achieve with the pipelined processor? **(1-point)**

Sol: $1400/300 = 4.67$

d) Assume we have a program with 5 instructions. What is the speed-up for this program when it executes on pipelined version compared to single cycle/non-pipelined version? **(1-point)**

Sol: $\text{pipelined_cc} = 5 + \text{instr} - 1 = 5 + 5 - 1 = 9$

$\text{pipelined_time} = 9 * 300 = 2700$

$\text{non_pipelined_time} = 5 * 1400 = 7000$

$\text{speed_up} = \text{non_pipelined_time} / \text{pipelined_time} = 2.59$

e) Assume we have another program with 500 instructions. What is the speed-up for this program when it executes on pipelined version compared to single cycle/non-pipelined version? **(1-point)**

Sol: $\text{pipelined_cc} = 5 + \text{instr} - 1 = 5 + 500 - 1 = 504$

$\text{pipelined_time} = 504 * 300 = 151200$

$\text{non_pipelined_time} = 500 * 1400 = 700000$

$\text{speed_up} = \text{non_pipelined_time} / \text{pipelined_time} = 4.62$

f) Are the speed-ups in part d and e same? If yes, why? If not, why? **(1-point)**

Sol: No, they are different. We need enough/large number of instructions to see pipeline gain/throughput.

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Q2: Assume that the following sequence of instructions is executed on a five-stage pipelined RISC-V processor.

```
ld x29, 40(x10)
and x18, x29, x16
sub x14, x29, x15
or x29, x29, x14
and x10, x16, x17
```

- a) If the processor has no forwarding or hazard detection unit, insert NOP(s) to ensure correct execution. Re-write the code sequence with NOP(s) inserted. **(5-points)**

Sol: ld x29, 40(x10)
 nop
 nop
 and x18, x29, x16
 sub x14, x29, x15
 nop
 nop
 or x29, x29, x14
 and x10, x16, x17

- b) Can we rearrange the original code to minimize the number of NOPs needed? If yes, re-organize and write the sequence of instructions. **(5-points)**

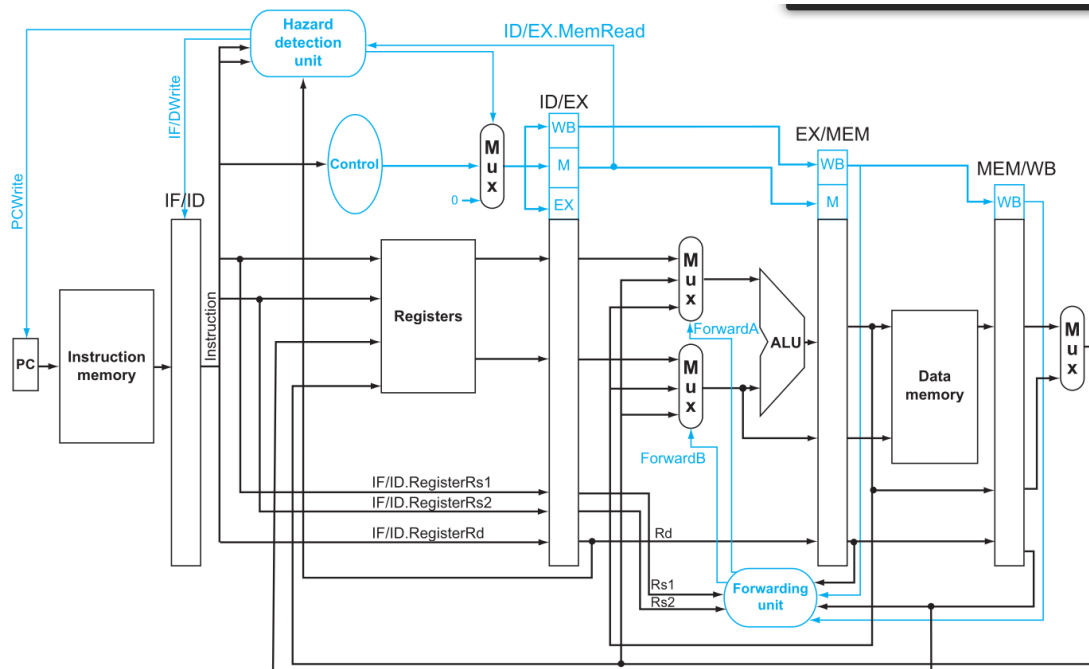
Sol: ld x29, 40(x10)
 and x10, x16, x17
 nop
 sub x14, x29, x15
 and x18, x29, x16
 nop
 or x29, x29, x14

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- c) Assume that the processor has both hazard detection unit and forwarding unit. Draw a multi-cycle diagram for the first ten cycles during the execution of original code and specify values of ForwardA, ForwardB, PCWrite, IF/IDWrite and Control MUX input for each cycle. PCWrite and IF/IDWrite are 0 when a hazard is detected, otherwise both signals are 1. Also, Control MUX value 0 means normal instruction, whereas 1 means hazard is detected. **(4-points)**

Instr	Cc0	Cc1	Cc2	Cc3	Cc4	Cc5	Cc6	Cc7	Cc8	Cc9
ld	F	D	E	M	W					
and → nop		F	D	-	-	-				
and			F	D	E	M	W			
sub				F	D	E	M	W		
or					F	D	E	M	W	
and						F	D	E	M	W
FwdA	-	-	00	00	01	00	00	00		
FwdB	-	-	00	00	00	00	10	00		
PCWrt	-	1	0	1	1	1	1	1		
IF/IDWrt	-	1	0	1	1	1	1	1		
Ctrl MUX	-	0	1	0	0	0	0	0		



Mux control	Source	Explanation
ForwardA = 00	ID/EX	The first ALU operand comes from the register file.
ForwardA = 10	EX/MEM	The first ALU operand is forwarded from the prior ALU result.
ForwardA = 01	MEM/WB	The first ALU operand is forwarded from data memory or an earlier ALU result.
ForwardB = 00	ID/EX	The second ALU operand comes from the register file.
ForwardB = 10	EX/MEM	The second ALU operand is forwarded from the prior ALU result.
ForwardB = 01	MEM/WB	The second ALU operand is forwarded from data memory or an earlier ALU result.